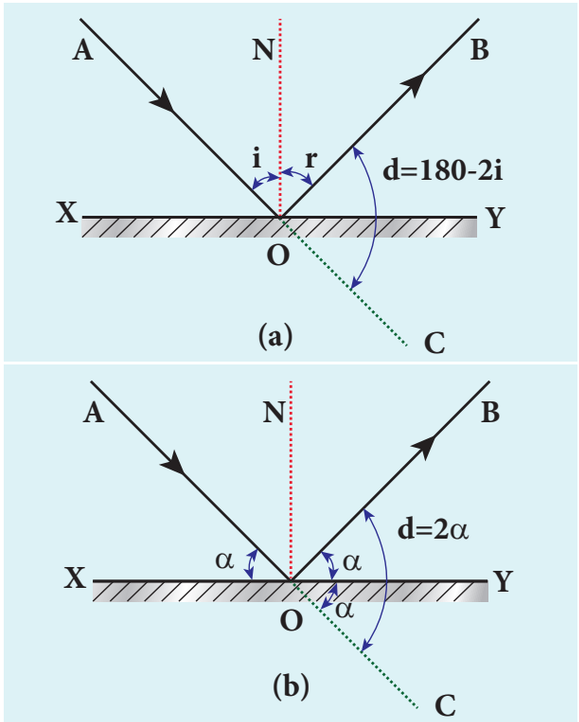


Figure 6.3 Angle of deviation due to reflection

EXAMPLE 6.1

Prove that for the same incident light when a reflecting surface is tilted by an angle θ, the reflected light will be tilted by an angle 2θ.

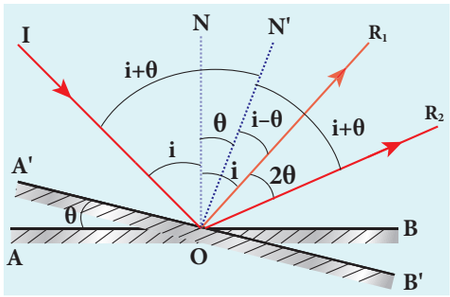
Solution

AB is the reflecting surface as shown in the Figure. Both the incident ray IO and the reflected ray OR_1 subtend angle i with the normal N as the angle of incidence is equal to angle of reflection. When the surface AB is tilted to $A'B'$ by an angle θ, the normal N is also is tilted to N' by the same angle θ. Remember that the position of the incident ray IO remains unaltered. But the reflected ray now is OR_2.

Now, in the tilted system, the angle of incidence, $\angle N'OI = i+\theta$ and the angle of reflection, $\angle N'OR_2 = i+\theta$ are the same. The angle between ON' and OR_1 is, $\angle N'OR_1 = i - \theta$. The angle tilted on the reflected light is the angle between OR_1 and OR_2 which is $\angle R_1OR_2$. From the geometry we can write,

$\angle R_1OR_2 = \angle N'OR_2 - \angle N'OR_1 = (i+\theta) - (i-\theta)$

$\angle R_1OR_2 = 2\theta$.



6.1.4 Image formation in plane mirror

Let us consider a point object A placed in front of a plane mirror. The point of incidence is O on the mirror as shown in the Figure. 6.4(a).

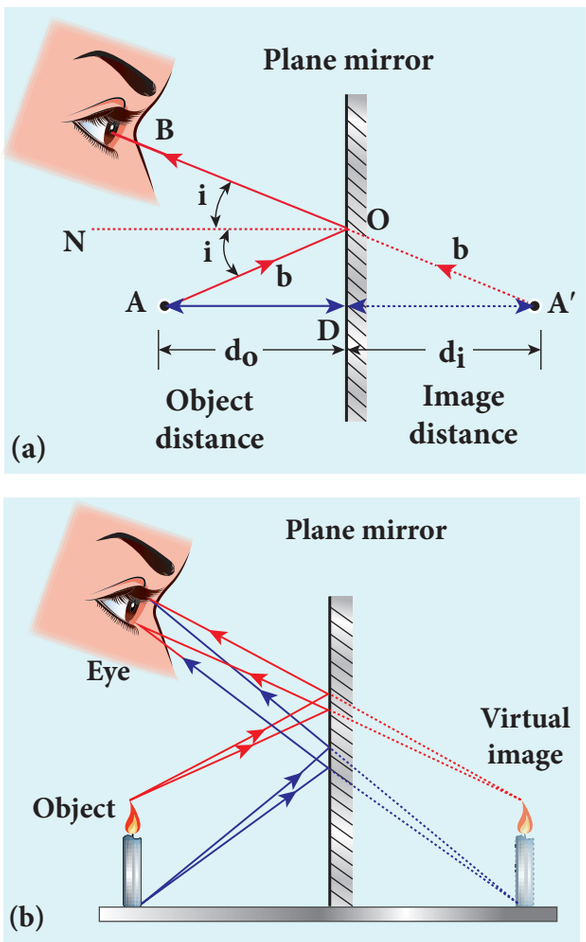


Figure 6.4 Formation of image in plane mirror for (a) point and (b) extended objects

UNIT-6(XII-Physics_Vol-2)_new.indd 3 21-07-2020 16:46:27